

数据格式转化nc_numpy

数据预处理

由于运行大模型，通常所需的格式为.npy的形式，但是ERA5给出的是 .nc 的形式，所以考虑将其转化为 .npy 的格式。

需要通过xaary来处理，为此，我们需要安装netCDF4的库

```
conda install -c conda-forge netcdf4
```

```
conda install -c conda-forge h5py
```

```
conda install xarray
```

为了明确.nc文件中，每一个变量的名字，可以通过Matlab的ncdisp库进行查询

```
ncdisp('E:\Project_Collection\2025\WEATHER\PANGU\DATA\6h\era5_surface_2025_05.nc')
```

检索发现，包含的4个变量名分别为

```
# 使用正确的变量名称（基于 ncdisp 输出）
msl = df_surface['msl'] # 海平面气压
u10 = df_surface['u10'] # 10米 U 风分量
v10 = df_surface['v10'] # 10米 V 风分量
t2m = df_surface['t2m'] # 2米温度
```

转化实例

以地表数据处理模式为例（具体的代码请见nc_npy）已给出详细的批注，只需要将其替换为本地的路径即可。

1.读取文件

```
surface_nc_file = r"此处替换为完整的实际地址"
```

2.设定存储文件地址

```
base_project_dir =  
r"E:\Project_Collection\2025\WEATHER\PANGU\Pangu_6h_Projec  
t" # 建议的基础项目目录  
save_dir_surface_npy = os.path.join(base_project_dir,  
"processed_npy_data", "surface_data_6h")
```

3.如果目录不存在则创建目录

```
os.makedirs(save_dir_surface_npy, exist_ok=True)
```

4.提取数据集的时间序

```
df_surface = xr.open_dataset(surface_nc_file)  
time_list_surface = np.array(df_surface.valid_time)
```

5.以循环的方式读取df_surface的四个变量

- 对于每一个时间点，从NetCDF文件中提取四个变量：平均海平面气压（msl）、10米U风分量（u10）、10米V风分量（v10）和2米温度（t2m）。
- 使用 `np.expand_dims()` 将每个变量的二维数组扩展为三维数组，形状为(1, 纬度, 经度)。

```

for i in range(len(time_list_surface)):
    time = time_list_surface[i]
    # 确保变量名与您的NetCDF文件中的名称匹配
    # 注意: ERA5通常使用'msl' (平均海平面气压), 'u10' (10米U风分量), 'v10' (10米V风分量), 't2m' (2米温度)
    msl =
np.expand_dims(df_surface.msl.sel(valid_time=time), 0)
    u10 =
np.expand_dims(df_surface.u10.sel(valid_time=time), 0)
    v10 =
np.expand_dims(df_surface.v10.sel(valid_time=time), 0)
    t2m =
np.expand_dims(df_surface.t2m.sel(valid_time=time), 0)

    res_array = np.concatenate([msl, u10, v10, t2m],
axis=0)

```

6.保存数据集并打印日志

```

# 将 numpy.datetime64 转换为类似 'YYYY-MM-DDTHH' 的字符串
time_str = np.datetime_as_string(time,
unit='h').replace(':', '').replace('-', '')
strname = f'mslp-10mU-10mV-2mT-{time_str}.npy'
save_path = os.path.join(save_dir_surface_npy, strname)
np.save(save_path, res_array.astype(np.float32)) # Ensure float32
print(f"{strname} saved successfully to {save_dir_surface_npy}")

```

执行完成后，可以得到如下图所示的执行过程

```
D:\Anaconda\envs\AI-Foundation\python.exe E:\Project_Collection\2025\WEATHER\PANGU\DATA\6h
Processing surface data from: E:\Project_Collection\2025\WEATHER\PANGU\DATA\6h\nc\era5_sur
mslp-10mU-10mV-2mT-20250601T00.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250601T06.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250601T12.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250601T18.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250602T00.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250602T06.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250602T12.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250602T18.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250603T00.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250603T06.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250603T12.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250603T18.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250604T00.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250604T06.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250604T12.npy saved successfully to E:\Project_Collection\2025\WEATHE
mslp-10mU-10mV-2mT-20250604T18.npy saved successfully to E:\Project_Collection\2025\WEATHE

Surface data processing complete.

Processing upper-air data from: E:\Project_Collection\2025\WEATHER\PANGU\DATA\6h\nc\era5_u
z-q-t-u-v-20250601T00.npy saved successfully to E:\Project_Collection\2025\WEATHER\PANGU\D
z-q-t-u-v-20250601T06.npy saved successfully to E:\Project_Collection\2025\WEATHER\PANGU\D
z-q-t-u-v-20250601T12.npy saved successfully to E:\Project_Collection\2025\WEATHER\PANGU\D
z-q-t-u-v-20250601T18.npy saved successfully to E:\Project_Collection\2025\WEATHER\PANGU\D
z-q-t-u-v-20250602T00.npy saved successfully to E:\Project_Collection\2025\WEATHER\PANGU\D
```